

IT7075C: Applied Artificial Intelligence for Cybersecurity Syllabus

Online — asynchronous delivery via Canvas (no scheduled class meeting time or place).

Instructor

Chengcheng Li

[Instructor email — TBD] · [Phone — TBD]

[Office location — TBD]

[Office hours — TBD (virtual, by appointment via Zoom)]

Teaching Assistant

[Teaching Assistant — none assigned / TBD]

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Course Description

This course explores how artificial intelligence (AI) enhances modern defensive cybersecurity practices. Students learn to apply AI-driven tools for identifying vulnerabilities, detecting attacks, modeling threats, and automating security operations. Emphasis is placed on using machine learning, deep learning, and Explainable AI to strengthen monitoring, incident response, and digital forensics. Learners gain hands-on experience with real cybersecurity datasets, assessment frameworks, and tools. By the end of the course, students will be able to design, evaluate, and implement AI-enabled cybersecurity solutions.

Pre-Reqs

None

Credit Hours and Course Format

3 credit hours; online (asynchronous)

Course Learning Outcomes

Upon completion of this course, the student will be able to:

1. Conduct security assessments using AI-enhanced tools and methodologies.
2. Perform threat modeling and map attack surfaces using AI-assisted techniques.
3. Develop AI and machine learning models to identify cyber risks and support mitigation.
4. Evaluate vulnerabilities and design AI-driven mitigation strategies.
5. Implement AI techniques to automate and improve security operations.
6. Produce clear reports with analysis and actionable recommendations.
7. Apply frameworks for AI-based threat intelligence.
8. Explain and apply principles of Explainable AI in cybersecurity.

Course Expectations

This is a hands-on, project-driven course. Each week pairs short concept lectures with runnable Jupyter notebooks that follow a slide → instructor-script → runnable-code rhythm, so you should expect to read, run, and modify real code every week. My teaching philosophy is that you learn applied AI for cybersecurity by building—and safely breaking—systems in an authorized lab environment, not by memorizing tools. You are expected to engage with the weekly materials, complete the six mini-projects and the final project, and treat all offensive or dual-use techniques as strictly limited to isolated lab VMs, instructor-approved or explicitly authorized targets, student-created apps, and public datasets. Tips for success: start early, run every notebook rather than only reading it, keep your API keys out of any shared or public repository, and post questions to the discussion board as soon as you get stuck. This course prepares you for roles in security engineering, detection and

response, and AI-security research by giving you working experience with LLMs, RAG, agentic frameworks (LangChain and LangGraph), the Model Context Protocol, and the orchestration of AI-driven security tooling under human control.

Student Support Statement

Welcome to the course! In SoIT, we have core values that we try to integrate into all teaching and learning in SoIT. We strive to create a safe, affirming environment for students where everyone has an equal opportunity to grow, flourish, and contribute. We strive to nurture independent thinking and expression of ideas while extending respect and civility to others in our thinking and actions. We strive to promote common interests while valuing individual differences. We strive to accept responsibility of our own commitment to excellence in these values and this learning environment. Your success in this class is important to me. We all learn differently and bring different strengths and needs to the class. If there are aspects of the course you're struggling with, please let me know and, together, we'll develop strategies to meet your needs and the requirements of the course.

Course Materials

All course materials are on Canvas.

Textbook

Required: Omar Santos, *Agentic AI for Cybersecurity: Building Autonomous Defenders and Adversaries* (Addison-Wesley, 2026). ISBN-13: 978-0-13-558986-1. This text anchors the RAG, LangChain, LangGraph, MCP, and Cybersecurity Orchestration modules (Chs. 1, 2, 4, 7, and 8).

Required Technology

Students will use Python 3.10+ with Jupyter notebooks (in VS Code, JupyterLab, or Google Colab) and Git/GitHub for version control. Core libraries introduced and used across the modules include NumPy, pandas, and Matplotlib; LangChain, LangGraph, and the Model Context Protocol (mcp); ChromaDB and FAISS for vector search; sentence-transformers for embeddings; Ollama for locally hosted open-weight models; and the OpenAI and Anthropic Python SDKs. Each module ships its own requirements.txt. You should already be comfortable with basic Python programming and command-line/Git usage. You will need an account and API key for at least one hosted LLM provider (OpenAI and/or Anthropic)—every notebook that needs a key is guarded, so the offline material still runs without one—and access to an isolated lab environment (for example, a local VM such as Metasploitable) for the orchestration/pen-testing module. A Google account is recommended for optional GPU work in Google Colab.

Recommended Resources

Video: Omar Santos, RAG, AI Apps, and AI Agents for Cybersecurity and Networking (LinkedIn Learning). · Companion code: github.com/santosomar/ai-agents-for-cybersecurity. · Frameworks referenced throughout the course: NIST AI Risk Management Framework (AI RMF), MITRE ATLAS, and the OWASP Top 10 for LLM Applications.

Course Policies and Grading

Attendance/Participation Expectations

Because this course is fully online and asynchronous, “attendance” is measured by your weekly engagement in Canvas: working through the module notebooks, submitting assignments by their deadlines, and participating in the discussion forums. There are no live meetings. I expect regular, substantive interaction—most weeks include a discussion forum where you respond to a prompt and reply to peers, and several projects involve collaborative problem-solving. To meet the course outcomes, log in to Canvas several times per week, keep pace with the weekly module schedule, run and experiment with every notebook, and post questions to the discussion board so the whole class benefits.

Absences

In an asynchronous course there is no class meeting to miss, but falling behind on the weekly schedule is the equivalent risk. You may work ahead within a module, but all assignments are due by the dates listed in the Course Schedule. If a personal, medical, or professional emergency will keep you from staying on pace for several days, contact me as early as possible and we will develop a written plan to complete the missed work. Extended, uncommunicated inactivity will affect your participation grade and may jeopardize your standing in the course.

Group Work Expectations

The six mini-projects are completed individually. The final project—the AI security analyst for your own lab—is a two-person capstone; you may petition to work alone, but pairs are the default because the build spans a tool layer and an assistant layer. When group work is used, each member is expected to contribute equally, and you may be asked to submit a brief peer evaluation. For collaborative work in this online setting, coordinate through Canvas groups, a shared Git repository, and video calls (Zoom or Teams). Regardless of grouping, every student must be able to explain and reproduce their submitted work independently.

Generative AI Policy

This is an applied-AI course, so generative-AI tools are part of the subject matter and are generally permitted—with clear expectations: [this guide](#)

- Generative AI (ChatGPT, Claude, GitHub Copilot, etc.) is permitted and encouraged as a learning and coding aid, provided you understand, can explain, and take full responsibility for everything you submit.
- Allowable use varies by assignment type: for coding and mini-projects you may use AI to draft, debug, and explain code; for reflective, analysis, or exam-style work, submitting AI-generated prose as your own is not allowed unless the assignment explicitly states otherwise. When AI meaningfully contributes to a deliverable, briefly disclose how you used it.
- Know the limitations in this domain: LLMs hallucinate CVEs, APIs, and command syntax, can produce insecure or subtly incorrect code, and are sometimes confidently wrong about security facts—always verify against authoritative sources and by running code in a safe environment. Never paste secrets, API keys, or private/authorized-target data into a hosted AI tool.
- Unauthorized or undisclosed AI use—submitting AI output as your own where it is prohibited, or misrepresenting AI-assisted work—is treated as an academic-integrity violation under UC policy, with consequences ranging from a zero on the assignment to a formal report to the University.

Communication Policy

The best ways to reach me are Canvas messaging and my UC email; I typically respond within 24 hours on weekdays and within 48 hours on weekends and holidays. Course-wide questions are best posted to the Q&A discussion forum so that everyone benefits from the answer. I post announcements in Canvas regularly—please enable Canvas notifications so you receive them by email and/or text, since announcements often contain schedule updates and assignment clarifications for which you are responsible.

Feedback

I aim to grade and return feedback on assignments and mini-projects within one week of the due date. Most assignments are graded with rubrics posted in Canvas alongside the assignment, so you can see exactly how points are allocated before you submit. For code-based projects, feedback will reference specific parts of your notebook or repository. If you have a question about a grade, please reach out within one week of receiving it.

Assignments and Submission Expectations

This course uses hands-on assignments rather than traditional exams: an opening lab, six mini-projects (set up your AI development environment; compare LLM hosting across local, Colab, and a cloud GPU; a framework-free RAG pipeline; a LangChain RAG application; a LangGraph agent; and an MCP server/client), discussion-forum posts, and a final project in which you build a scoped scanning toolkit and a grounded AI security analyst for your own isolated lab. Every project uses the same competency-based shape—a small, bounded build, the evidence it produces, and a demonstration and analysis delivered as either a 2-

page report or a 5–7 minute video, with the analysis carrying most of the credit. Unless stated otherwise, submit code as runnable Jupyter notebooks (.ipynb) and/or a linked Git repository, plus a short written report where required, through Canvas. Written work should be clear, well-organized, and professional; grammar, spelling, and professionalism are part of the grade on reports. When you cite sources or datasets, use a consistent citation style (APA or ACM); I will point you to resources, but you are responsible for correct citation.

Late or Missing Assignment Policy

Late assignments lose 10% per day (rounded up) for up to three days; after three days late, work is not accepted and receives a zero unless you arranged an extension with me in advance. If an emergency arises, contact me before the deadline whenever possible so we can make a plan. Because this is an online course in which you may work ahead, plan for technology and life disruptions. Selected assignments may be revised and resubmitted for partial credit when the assignment instructions explicitly allow it.

Grading Scale

93.00% and above = A
90.00%-92.99% = A-
87.00%-89.99% = B+
83.00%-86.99% = B
80.00%-82.99% = B-
77.00%-79.99% = C+
73.00%-76.99% = C
70.00%-72.99% = C-
67.00%-69.99% = D+
63.00%-66.99% = D
60.00%-62.99% = D-
less than 60.00% = F

The course grades are weighted as follows:

- Grades are based on weighted categories. The six mini-projects together are worth 60% of the grade (100 points each), the final project is worth 20%, the Module 1 opening lab is worth 5%, discussion-forum participation is worth 10%, and overall participation and engagement is worth 5%. The table below shows the point allocation (1,000 points total).

Assignments	Points Assigned	Weight
Mini-Projects (MP1–MP6)	600 (100 each)	60%
Final Project — AI Security Analyst	200	20%
Discussion Forums	100	10%

Assignments	Points Assigned	Weight
Module 1 Lab (AI landscape)	50	5%
Participation & Engagement	50	5%

Course Schedule

The schedule below lists the weekly modules, topics, and major deliverables with due dates. Dates follow the University of Cincinnati Fall 2026 academic calendar; please verify against the official UC calendar and Canvas, as dates may shift. Each week's materials, detailed instructions, and rubrics are posted in the corresponding Canvas module.

Week	Dates	Modules/Topics	Assignments/Due Dates
1	Aug 24	Module 1 — Introduction to AI & the AI-for-Cybersecurity Landscape	Module 1 Lab due; accounts & API-key check
2	Aug 31	Module 2 — Tools & Environment (VS Code, Python, Git, notebooks, secrets)	MP1 — AI Tools & Environment Setup due
3	Sep 7	Module 3 — LLMs on Local & Cloud (Labor Day 9/7)	MP2 — Running LLMs in Different Environments due
4	Sep 14	Module 4 — Retrieval-Augmented Generation (RAG), Part 1: embeddings & vector DBs	Discussion 1
5	Sep 21	Module 4 — RAG, Part 2: the retrieve → augment → generate pipeline	MP3 — Build & Tune a RAG Analyst due
6	Sep 28	Module 5 — LangChain, Part 1: chains, prompt templates, LCEL	Discussion 2
7	Oct 5	Module 5 — LangChain, Part 2: RAG with a framework	MP4 — LangChain RAG Application due
8	Oct 12	Module 6 — LangGraph, Part 1: state, nodes, edges	Discussion 3
9	Oct 19	Module 6 — LangGraph, Part 2: stateful, tool-using agents	MP5 — LangGraph Security Agent due

Week	Dates	Modules/Topics	Assignments/Due Dates
10	Oct 26	Module 7 — Model Context Protocol (MCP), Part 1: the protocol & servers	Discussion 4
11	Nov 2	Module 7 — MCP, Part 2: client/agent + MCP as an attack surface	MP6 — MCP Server & Client due
12	Nov 9	Module 8 — Cybersecurity Orchestration, Part 1: scoped tooling & simulated recon	Final project proposal & pair sign-up due; Discussion 5
13	Nov 16	Module 8 — Cybersecurity Orchestration, Part 2: chain vs. retrieval vs. agent	Final project checkpoint — lab & tools working
14	Nov 23	Final project build week (Thanksgiving 11/26–27)	Discussion 6
15	Nov 30	Final project build week; demonstrations & analysis	Final project report/video due
Finals	Dec 7	Final demonstrations & course reflections	All final project deliverables due

UC Policies

Attendance: Students at the University of Cincinnati are expected to attend classes in order to meet the learning objectives for the course. Students are expected to follow the class attendance policy as outlined by the instructor. Students participating in official University sponsored activities, where the student is representing the University, will meet with instructor related to absences due to university-sponsored activities. When such absences occur, students need to provide documentation of the event(s), and work with the professor to develop a written plan for completion of missed assignments or other course requirements with as much advance notice as possible.

Communication: All university business must be conducted *using a university assigned email account*. The use of a personal, or non-university assigned email account to conduct university business is strictly prohibited. [See full policy here](#).

Class Cancellation Policy: In the rare case that a class must be cancelled, faculty will post an announcement on the LMS that will be emailed to students. Faculty will attempt to communicate class cancellations with as much advance notice as possible. Students should be sure that their LMS email is current and valid to ensure emails are received. If the University closes due to inclement weather or other emergency situations, there will be an announcement posted on Canvas (LMS) and if possible on the local news channels (TV, apps, news websites). Communications related to University closures will also be sent to the student's cell phone number on record through the automatic University emergency text messaging system. Students should notify the University if they change their cell phone number to ensure they will receive these important emergency communications.

Academic Integrity: The University of Cincinnati is committed to academic integrity. A formal articulation of this commitment is articulated in the [Student Code of Conduct](#). The University Rules, including the Student Code of Conduct, and other documented policies of the department, college, and university related to academic integrity will be enforced. Any [violation of these regulations](#), including acts of plagiarism or cheating, will be dealt with on an individual basis according to the severity of the misconduct. At the least, you should anticipate a zero on an assignment, at the fullest extent, your violation will be reported to the university and the incident will be documented on your permanent record.

Accessibility and Special Needs: If you have a qualifying disability, UC's [Accessibility Resources](#) and your college's Access Coordinator are here to ensure you receive reasonable accommodations. Please do this as soon as possible so AR can work with me to ensure you have full and equal access to our learning environment.

I encourage you to contact [Accessibility Resources](#) to schedule a confidential meeting to discuss services and accommodations, or to begin by [registering for accommodations](#). UC uses an online system called [UC Accommodate](#), where you'll be able to register and manage your accommodations (use your UC username and password for access). Below are a few helpful UC Accommodate resources:

- [UC Accommodate Student Portal Overview](#)
- [How to Schedule an Appointment in UC Accommodate](#)
- [How to Submit a Semester Request in UC Accommodate](#)

Counseling Services: As a UC college student, there may be times when personal stressors interfere with your academic performance and/or negatively impact your daily life. If you or someone you know is experiencing challenges such as strained relationships, increased anxiety, feeling down, difficulty concentrating, lack of motivation, etc., you are not alone; there are people here that can help.

- **To schedule:** please contact Counseling and Psychological Services (CAPS) at **513-556-0648** to schedule a virtual or in-person appointment with our CAPS CECH

embedded counselor, Madeleine Muller, located in the CECH Student Services Center in Suite 360 of the Teachers-Dyer Complex.

- Have questions before committing to an appointment? Check out Let's Talk, a 15-20-minute conversation to ask questions about counseling and get to know our [CECH counselor](#).
- **After-hours crisis line: 513-556-0648**
- **Self-Help:** to access Therapy Assistance Online (TAO), a **free** [self-help resource](#) for students.

Religious Accommodation: Ohio law and the University's Student Religious Accommodations for Courses Policy 1.3.7 permits a student, upon request, to be absent for reasons of faith or religious or spiritual belief system or participate in organized activities conducted under the auspices of a religious denomination, church, or other religious or spiritual organization and/or to receive alternative accommodations with regard to examinations and other course requirements due to an absence permitted for the above-described reasons. Not later than fourteen days after the first day of instruction in the course, a student should provide the instructor with written notice of the specific dates for which the student requests alternative accommodations. For additional information about this policy, please contact the Executive Director of the Office of Equal Opportunity and Access at (513) 556-5503 or oeohelp@UCMAIL.UC.EDU

Title IX: The University of Cincinnati is committed to providing a living, learning, and working environment that is free from discrimination, harassment, and retaliation. Title IX of the Education Amendments of 1972 is a federal law that prohibits discrimination on the basis of sex in education programs or activities that receive federal financial assistance. Consistent with federal, state, and local laws and regulations, the University prohibits discrimination and harassment on the basis of sex, in its academic, admissions, education and employment programs and activities. Students who wish to know more about their rights and resources on campus, they can consult the [Title IX website](#) or contact the Title IX office directly at 513-556-3349.

NOTE: The instructor reserves the right to update this syllabus as class needs arise. Be assured that they will communicate to you any changes to the schedule, syllabus or policies quickly and efficiently.